



Bee TV

Interactive TV Series module

Architecture presentation · search · series details · watch tracking · comments · AI-powered Bee Review

Agenda

1. **Context and approach**: specification first, then principles, then decisions
2. **Product tour**
3. **Architectural solution**: tiers, backend slices, frontend features, vendor independence
4. **Information flow**: TV Series Search · Series Details (sequence diagrams)
5. **AI component**: design, runtime flow, fallback strategy
6. **Quality attributes**: resilience, error handling, persistence, testability, DevOps
7. **Trade-offs and roadmap**

Context and approach

- Early-stage streaming platform · **MVP** · the module is a **reference for future teams**
- **Specification first**, before any code:
 - `docs/specification.md`: product features and UI hints
 - `docs/technical-architecture.md`: the principles and drivers (the *macro-decisions*)
 - `docs/patterns-and-choices.md`: every implementation decision traced back to a driver
- Evaluation drivers kept in view the whole time:

Architecture & design	Code quality	Functionality & stability	AI integration	Testability
30%	20%	30%	10%	10%

Product tour: search



Bee TV

Find a series. Track every episode. Get the buzz.



What are we watching today?

Type the name of a series to explore its seasons, track episodes
and get a Bee Review.

© Bee TV. Made with honey.

Series data by [TVMaze](#) (CC BY-SA)

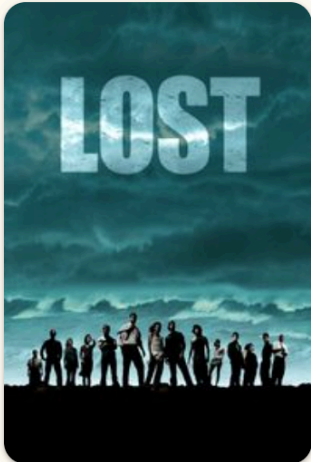
Search-engine-style home · results load asynchronously into a responsive tile grid (title, year, poster) · the query lives in the URL

Product tour: series details



Search TV series...

← Back to results for "lost"



✦ Bee Review

0 of 150 episodes watched

Lost

2004–2010 · ABC · Ended · English ★ 8.2

Out of the blackness, the first thing Jack senses is pain. Then burning sun. A Bamboo forest. Smoke. Screams. With a rush comes the horrible awareness that the plane he was on tore apart in mid-air and crashed on a Pacific island. From there it's a blur, as his doctor's instinct kicks in: people need his help. Stripped of everything, the 48 survivors scavenge what they can from the plane for their survival. Some panic. Some pin their hopes on rescue. A few find inner strength they never knew they had -- like Kate, who, with no medical training, suddenly finds herself suturing the doctor's wounds. Hurley - a man with a warm sense of humor despite the desperate situation - does his best to keep his cool as he helps those around him to survive. Charlie is a faded rock star who harbors a painful secret. Sayid is a Middle Eastern man who must wrestle with the racial profiling directed at him by some of his fellow survivors. Jin and Sun are a Korean couple whose traditions, values and language are foreign, and thus causes much to get lost in the translation. Sawyer has an air of danger surrounding him, and his intense sense of mistrust for everyone around him could prove to be fatal to his fellow castaways. Michael has just gained custody of his nine-year-old son, Walt, after the death of his ex-wife - they are a father and son who don't even know each other. Locke is a mysterious man who keeps to himself, and who harbors a deeper connection to the island than any of the others. Self-centered Shannon - who actually gives herself a pedicure amid the chaos - and her estranged controlling brother, Boone, constantly bicker and must learn to get along if they are to survive. And young Claire is eight months pregnant and ill-prepared for the hardships of motherhood - especially on a deserted island.

Drama

Adventure

Supernatural

Episodes

Season 1 (26)

Season 2 (27)

Season 3 (40)

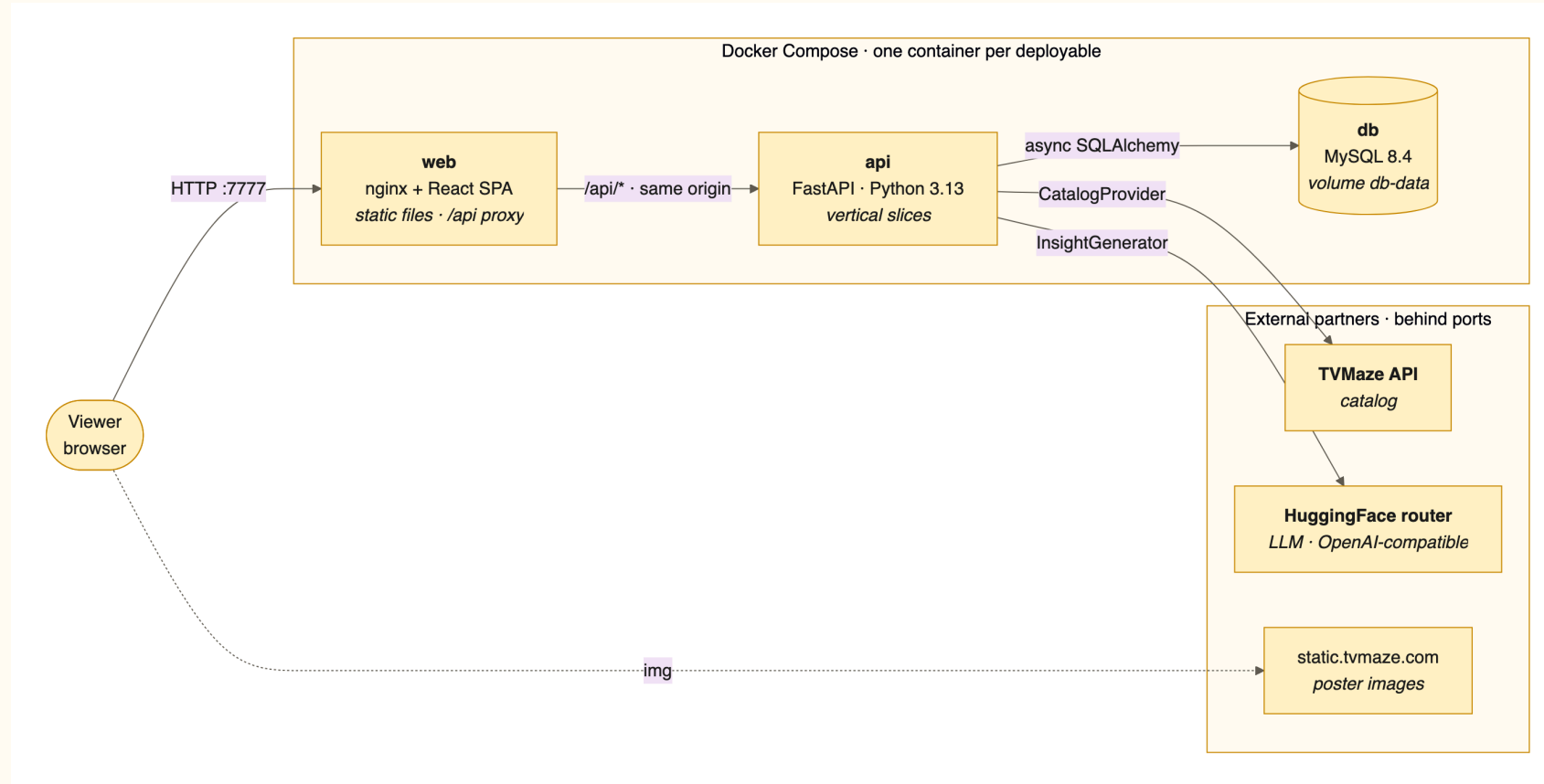
Season 4 (16)

Season 5 (20)

Season 6 (21)

Poster, summary and genres · episodes grouped by season · watched toggles · per-episode comments · **Bee Review** for the series or any episode

Architectural solution: two tiers, containerized

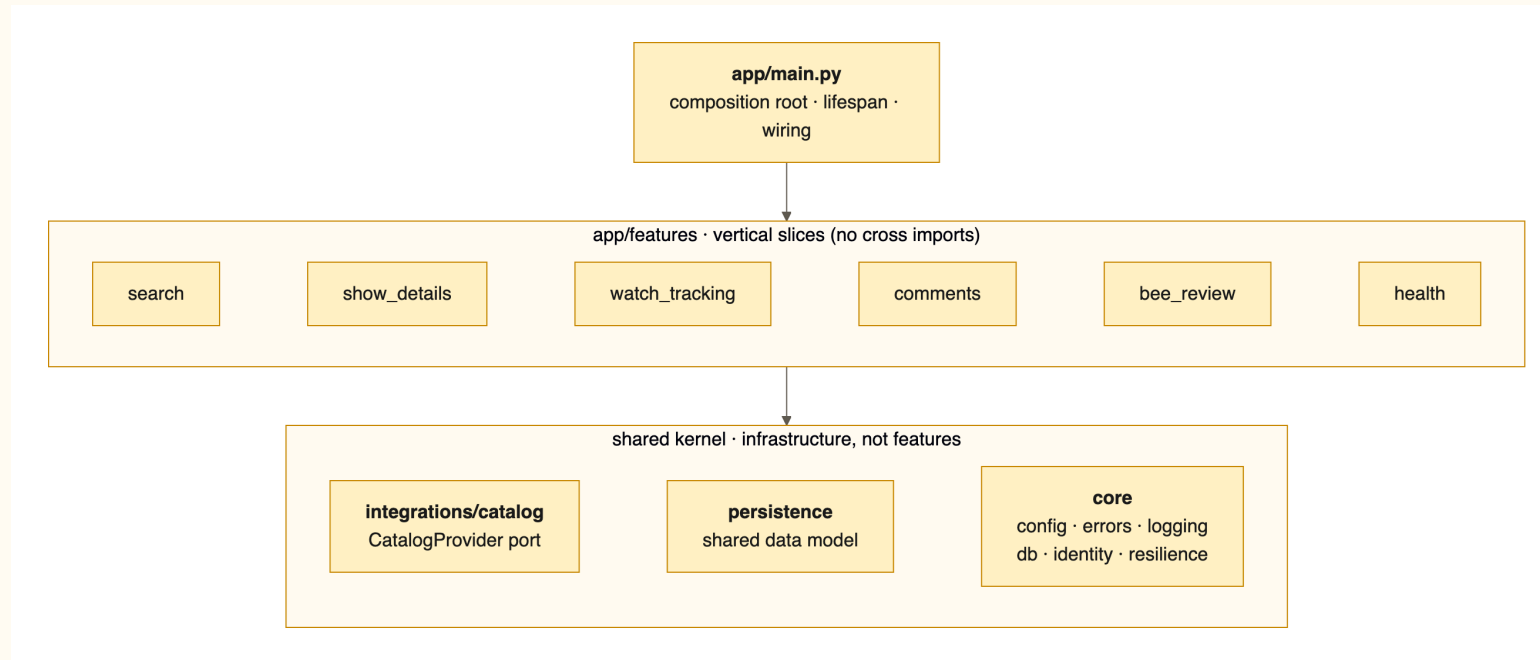


One container per deployable · the SPA and the API share **one origin** (nginx proxies **/api**) · partners are reached **only by the backend**, behind ports

From drivers to decisions

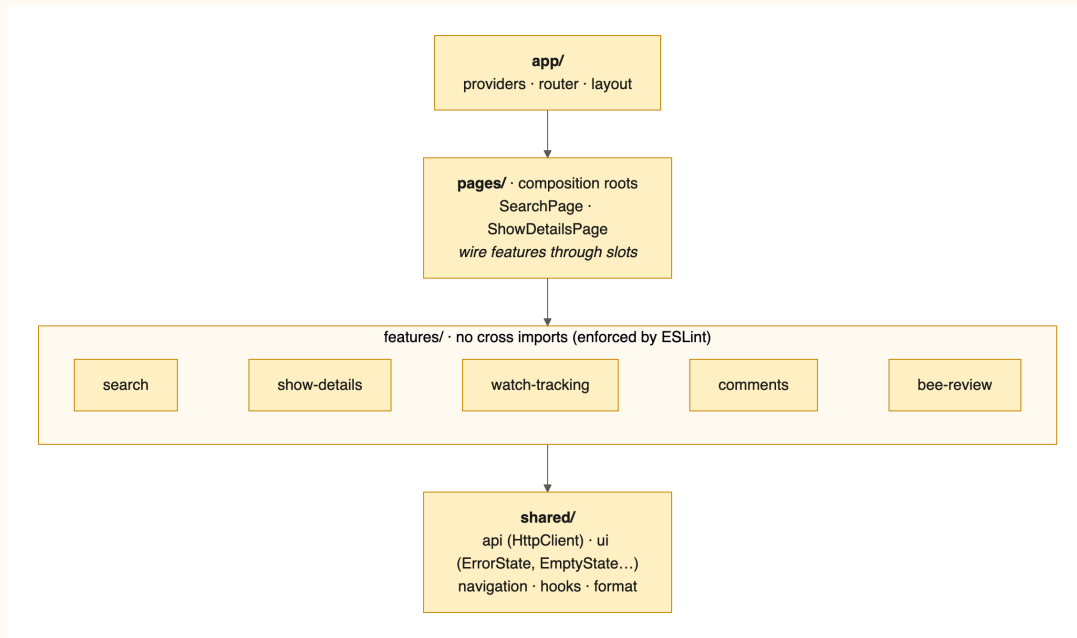
Driver (specification)	Key decisions
Vertical Slice Architecture, use cases in handlers	6 self-contained slices · shared kernel · transaction script per endpoint
SOLID · testable and mockable types	Ports (ABC) and adapters · DI via Depends · fakes over mocks · injectable clocks
Vendor lock-in proof	CatalogProvider anti-corruption layer · provider-agnostic models
Global exception handlers · per-integration retries	RFC 9457 problem details · retry, circuit breaker and cache chosen per partner
Feature folders · design system (MUI)	Lint-enforced boundaries · composition via slots · theme tokens
Friendly error UX for unavailability	Error taxonomy → mascot copy · partial degradation · optimistic rollback
Single command · tests gate every build	Compose 3.3 · multi-stage images with a test stage

Backend: Vertical Slice Architecture



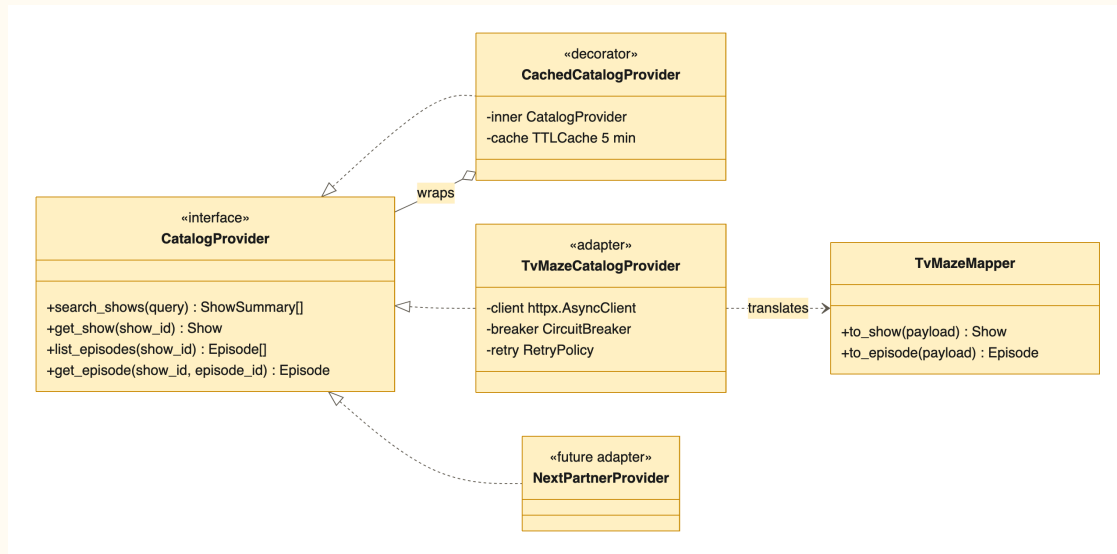
- **One folder per capability:** endpoints, DTOs, repository and helpers together; **no slice imports another**
- **Share the data model, not the code:** Bee Review reads comments through its own `OpinionSource`
- **Use case in the handler** (transaction script) · pure helpers for logic worth testing (`group_by_season`)

Frontend: feature folders, composed by pages



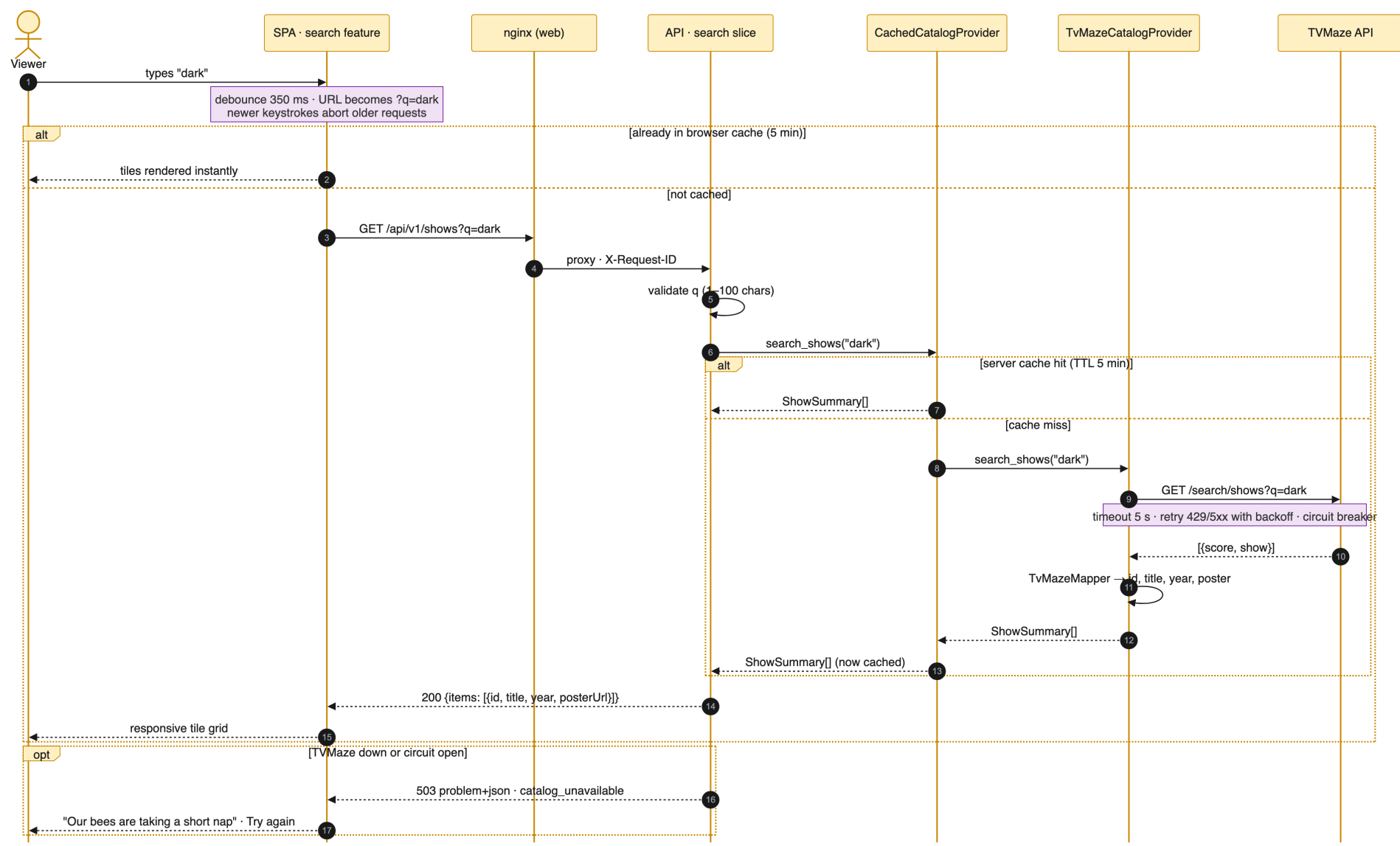
- **Layered rule:** `app` → `pages` → `features` → `shared`, never sideways
- **ESLint `no-restricted-imports`** fails the build on a cross-feature import
- **Inversion of Control via slots:** `EpisodeTile` exposes `renderTitleAdornment` and `renderActions`; the page injects the watched toggle, the comments button and Bee Review
- **DI through context:** features use the `HttpClient` interface, never `fetch`
- **Server state** via TanStack Query: key factories, cancellation, optimistic updates

Vendor lock-in proofing: catalog anti-corruption layer



- **Port:** **CatalogProvider**, 4 methods (ISP)
- **Adapter:** TVMaze HTTP · **mapper:** TVMaze JSON → Bee TV models
- **Decorator:** TTL cache added at composition time
- **Sanitized:** partner HTML → plain text (XSS-safe UI)
- **Defensive:** ids validated before calling; episode ownership checked
- A new partner = **one new adapter + one factory line**; slices are untouched

Information flow 1: TV Series Search



TV Series Search: what the flow guarantees

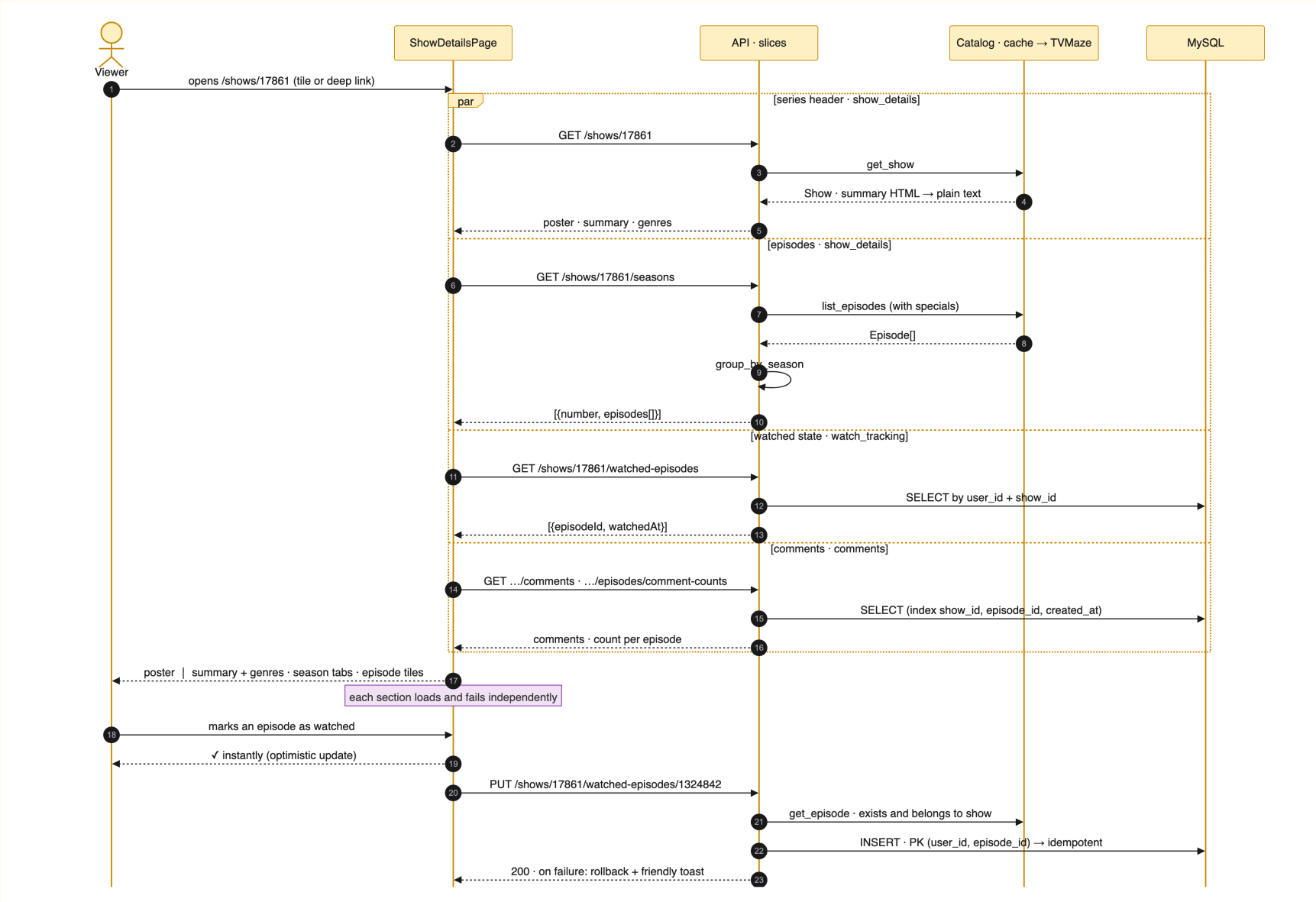
Responsiveness

- **Debounce 350 ms**, then the URL `?q=` (shareable, back/forward, bookmarkable)
- **Abort** of outdated requests (`AbortSignal`)
- **Two cache levels**: browser (TanStack Query) and server (TTL 5 min, normalized key)
- Previous results stay visible while refetching, with a thin progress bar

Robustness

- **Edge validation**: `q` 1–100 characters, otherwise `422`
- **TVMaze**: timeout 5 s · 3 attempts with exponential backoff (only 429/5xx/transport) · **circuit breaker**
- Outage: `503` `problem+json` → *"Our bees are taking a short nap"* + retry
- Responsive, accessible tile grid (`list` semantics, poster `alt`)

Information flow 2: Series Details



Series Details: what the flow guarantees

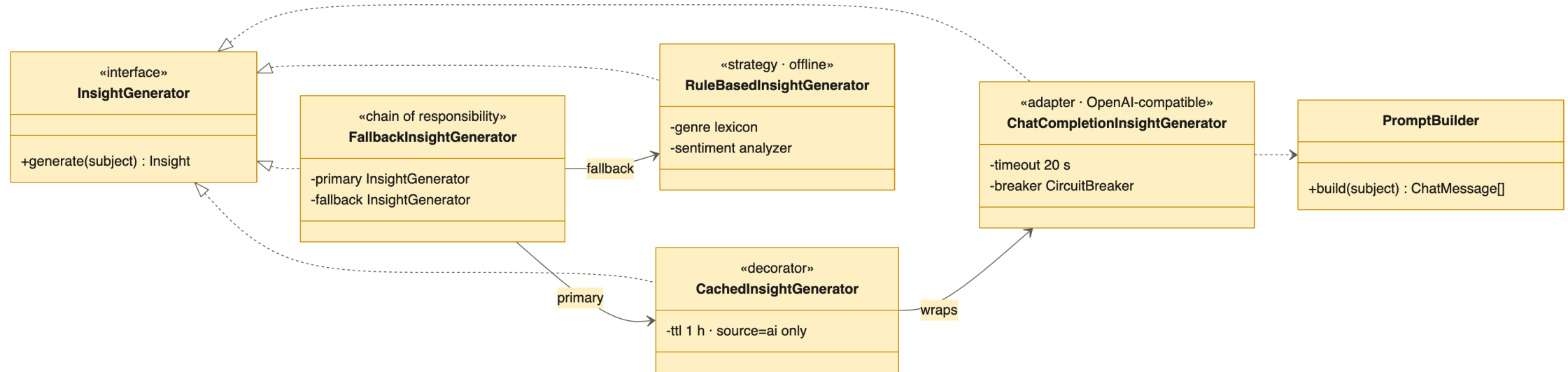
Composition and data

- **Four slices in parallel**: header, seasons, watched state, comments (with counts)
- `group_by_season`: seasons ascending, specials last, un-numbered episodes by air date
- **Summary HTML** → **plain text** on the server
- Dedicated URL `/shows/{id}` · deep links · a **back link** to the real previous in-app page

Robustness

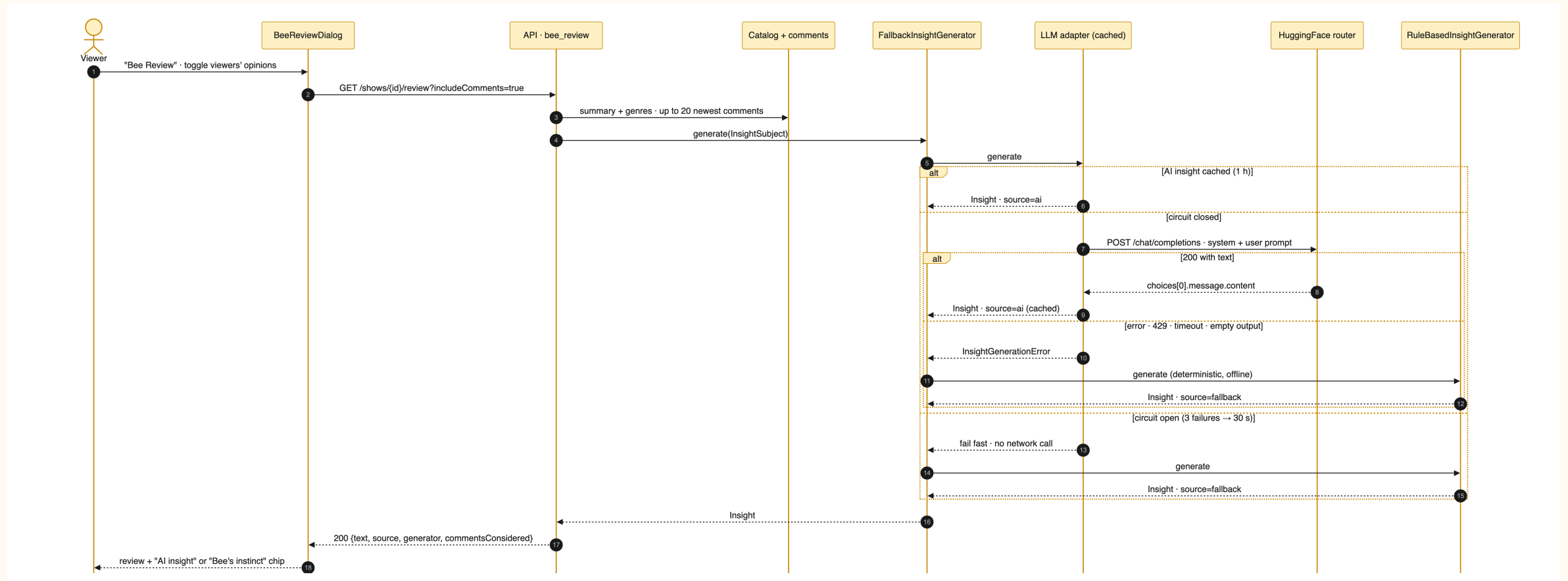
- **Sections fail independently**: skeletons and inline errors, not a blank page
- **Watched toggle**: optimistic ✓ · `PUT / DELETE` are **idempotent** · rollback + toast on error
- **Existence check** before writes (no orphan rows)
- Race-safe insert: composite PK `(user_id, episode_id)` + `IntegrityError` handling

AI component: design



- **Port** `InsightGenerator`: the rest of the system never knows a model exists
- **Adapter** speaks the **OpenAI-compatible** protocol: HuggingFace, OpenAI, Groq, Ollama or vLLM is a matter of **configuration**
- **Strategy** (LLM or rules) · **Chain of Responsibility** (fallback) · **Decorator** (cache for AI results only)
- **No token configured**: the composition root wires the rule-based strategy only

AI component: runtime flow and fallback strategy



No retries on the LLM (a fast fallback beats a slow retry) · circuit breaker 3 failures → 30 s · every response states its **source**

AI component: prompt, safety and transparency

Prompt engineering

- Role: *Bee, the friendly TV critic*
- Constraints: 2–3 sentences, at most 70 words, **only provided facts**
- One-shot tone example (from the brief)
- Inputs: title, genres, summary (≤ 1500 chars), **optional** viewer comments ($\leq 20 \times 300$ chars)

Safety

- Comments are **delimited and declared untrusted** (prompt-injection mitigation)
- Output normalized and capped (700 chars) · token secret never logged or sent to the browser

Fallback engine (offline, deterministic)

- Genre $\rightarrow$ phrase lexicon · first sentence of the summary
- Lexicon **sentiment analysis** of comments $\rightarrow$ "*viewers are mostly enthusiastic (3 comments considered)*"

Transparency in the UI

- **"AI insight"** chip vs **"Bee's instinct"** chip + caption
- Toggle "*Consider viewers' opinions*" $\rightarrow$ `includeComments=true`

Resilience: chosen per integration, by context

Integration	Timeout	Retry	Circuit breaker	Cache	Degraded mode
TVMaze	5 s	3×, backoff, 429/5xx only	3 → open 30 s (404 is success)	TTL 5 min	503 catalog_unavailable
LLM	20 s	none (UX, quota)	3 → open 30 s	AI results, 1 h	Rule-based insight, labeled
MySQL	driver	startup wait only (30 × 2 s)	n/a	n/a	503 database_unavailable · health degraded
SPA → API	browser	queries: 2× for offline/5xx · mutations never	n/a	query cache	Friendly error states

The circuit breaker **wraps** the retries: one logical call counts once, so a partner outage fails fast instead of piling up slow requests.

Error handling as architecture

Backend

- Exception **hierarchy with HTTP metadata** (status, code, title) → adding an error never touches a handler (OCP)
- **Global handlers**: application, validation, unknown routes, DB connection (→ 503), catch-all 500 that hides internals
- **Exception translation** at adapters: vendor errors never leak into slices
- **RFC 9457** application/problem+json

Frontend

- `classifyError()` → offline · unavailable · not-found · invalid · unknown
- Table-driven `ERROR_COPY`: mascot mood, human copy, retryable or not
- `ErrorState` full page / compact inline · toasts for mutations
- **Keep the user's work**: comment draft preserved, optimistic rollback

Persistence and schema deployment

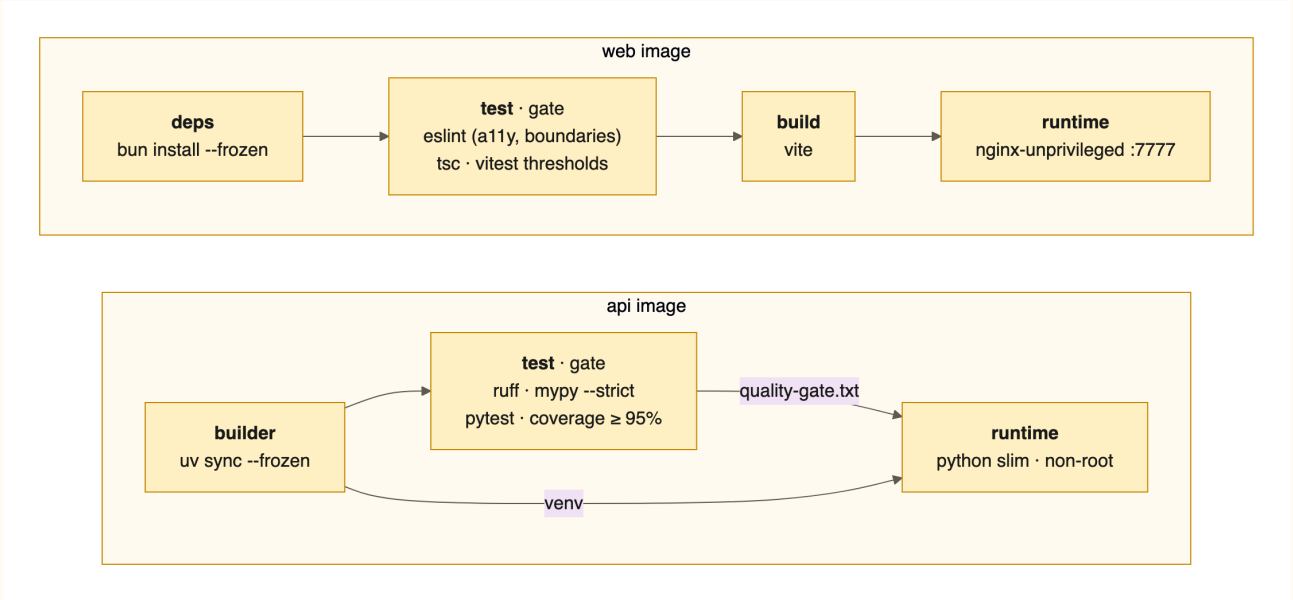
Schema (MySQL 8.4 · utf8mb4)

- `comments`: series or episode scope
(`episode_id` nullable) · index (`show_id`,
`episode_id`, `created_at`)
- `watched_episodes`: PK (`user_id`,
`episode_id`) → **idempotent** · index
(`user_id`, `show_id`)
- Ids as `VARCHAR(64)`: vendor-neutral, no foreign
keys to partner data
- `DATETIME(6)` in **UTC**

Deployment

- **Alembic migrations** = schema as code
- Container start: `python -m app.migrate` **waits
for MySQL**, then `upgrade head` (idempotent)
- Repository pattern per slice · session per
request
- Guest identity via **one dependency**:
authentication later touches one function

Testability and the mandatory quality gate



	Backend (API)	Frontend (App)
Tests	131 · pytest · unit + integration (ASGI, SQL, migrations)	122 · Vitest + Testing Library · unit + page flows
Coverage	≈ 99% (gate ≥ 95%)	≈ 99% lines (gate 95/95/95/90)
Static analysis	Ruff (incl. security rules) · mypy --strict	ESLint (typescript-eslint strict, a11y, boundaries) · tsc

No image can be built without passing the gate · fakes instead of mocks · injected clocks · partners faked, so the gate is deterministic

DevOps: everything with one command

```
docker compose up --build          # build → test gate → migrate → run · http://localhost:7777
```

- **Compose file format 3.3**: runs on legacy `docker-compose` v1 and on Compose v2/v5
- **Self-healing readiness**: API waits for MySQL; nginx resolves the API per request
- **Health checks** at every level: MySQL ping · `/api/health` (readiness) · `/healthz`
- **Multi-stage, non-root images** · frozen lockfiles · only port 7777 published
- **Zero configuration**: every variable has a default; `HF_TOKEN` is optional
- **Edge hardening**: CSP, security headers, immutable asset caching

Conscious trade-offs

Decision	Benefit	Cost / mitigation
In-process TTL caches	No Redis to operate	Per replica · swap via the decorator seam
Migrate on container start	Single command	Single replica only · <code>BEE_SKIP_MIGRATIONS</code> + release job to scale
Existence check before writes	No orphan rows	Writes return 503 during a catalog outage
No retries on the LLM	Fast, predictable UX	Transient blips fall back · AI cache recovers quickly
SQLite in the test gate	Fast, hermetic builds	MySQL specifics verified manually · candidate for a compose-level test
Mocked guest identity	MVP speed	<code>user_id</code> columns and one seam already in place

Roadmap

- **Identity:** real authentication behind `get_current_user` · per-user data already modeled
- **Scale:** shared cache (Redis) behind the existing decorators · migrations as a release job
- **Catalog:** pagination through the `Page` envelope · a second partner as a new adapter
- **AI:** provider choice per environment · streaming responses · evaluation set for prompt changes
- **Quality:** MySQL-backed integration tests in the pipeline · end-to-end browser tests
- **Operations:** structured JSON logs · metrics (latency, fallback rate, circuit state) · tracing

Thank you

Questions?

`docker compose up --build` → <http://localhost:7777>

Docs: [README.md](#) · [docs/technical-architecture.md](#) · [docs/patterns-and-choices.md](#)